BIG DATA

INFORMATION RETRIEVAL

SWIN BUR NE
SWINBURNE UNIVERSITY OF TECHNOLOGY

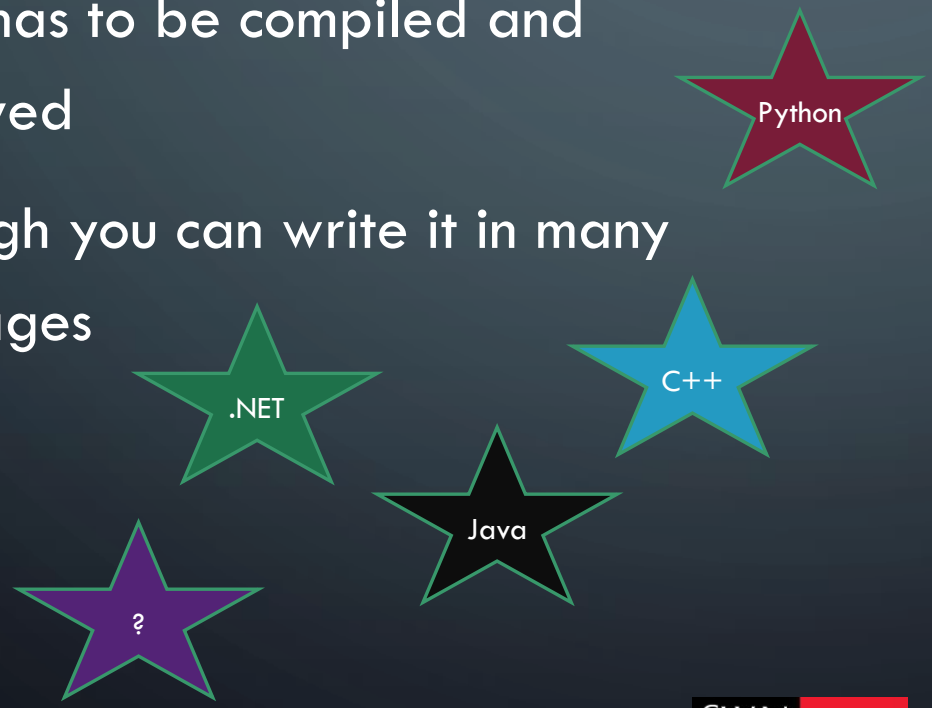
LEARNING OBJECTIVES

- At the end of this presentation, you should be able to
  - Explain how the Hadoop infrastructure facilitates Big Data processing;
  - Explain the role of Pig in making MapReduce processing more accessible;
  - Explain the role of Hive as an analytical tool for structured data;
  - Discuss how these technologies change the way we traditionally processed data.

SWINBURNE UNIVERSITY OF TECHNOLOGY

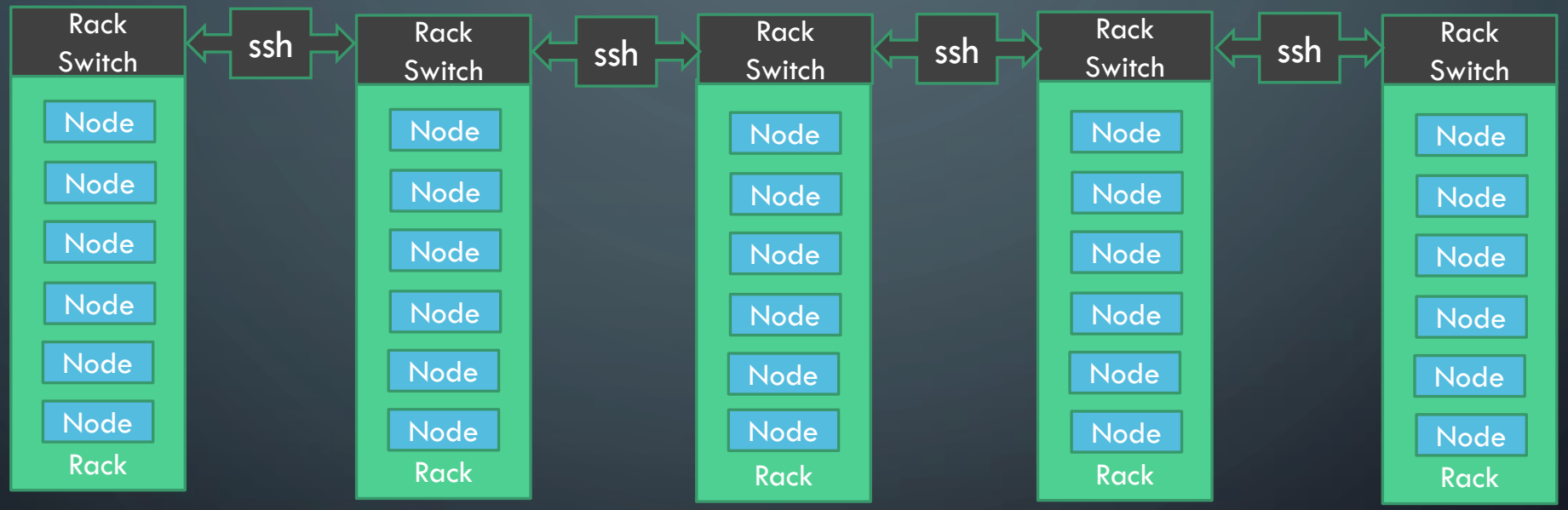
MAPREDUCE IS DIFFICULT

- MR is for experienced programmers
- Code has to be compiled and deployed
- Although you can write it in many languages



SWINBURNE UNIVERSITY OF TECHNOLOGY

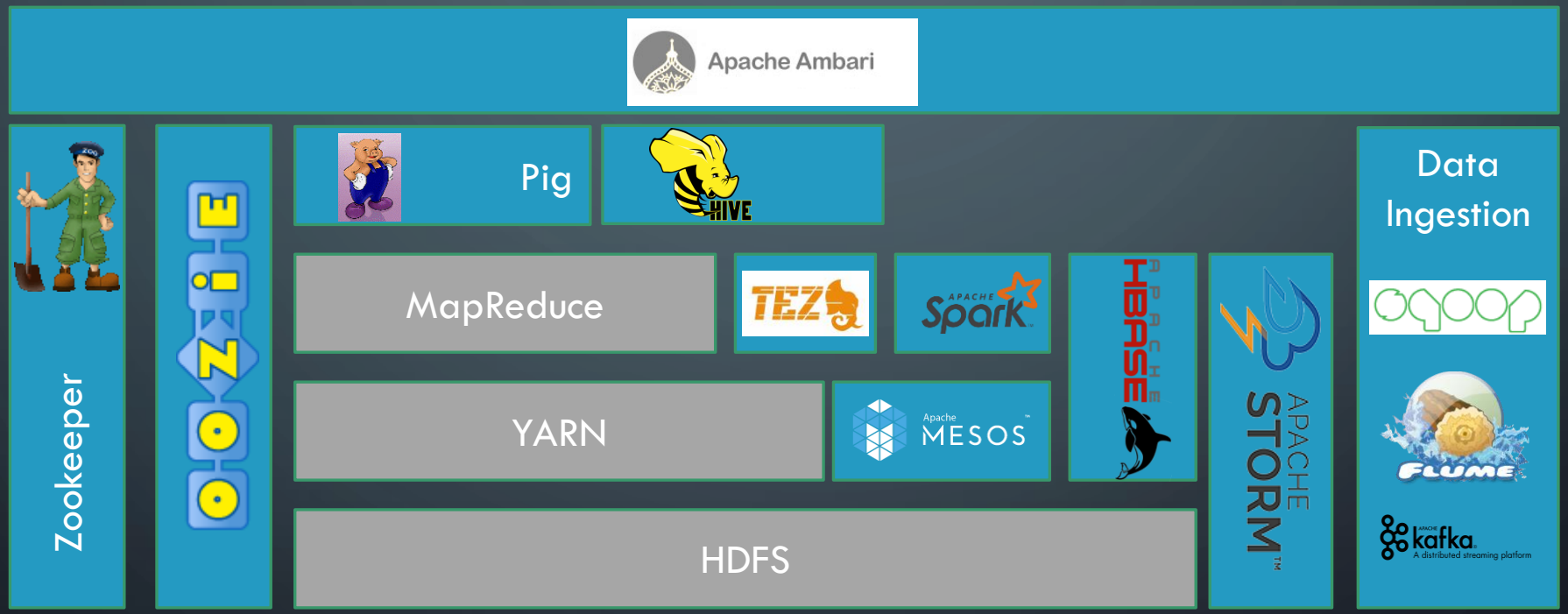
HADOOP CLUSTERS ARE GREAT



but MapReduce is complicated!

SWINBURNE UNIVERSITY OF TECHNOLOGY

HADOOP 'ECOSYSTEM'



SWINBURNE UNIVERSITY OF TECHNOLOGY

AND ALSO...

PIG

Hadoop Tool for unstructured data

SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG - MAPREDUCE

PIG

- High-level data processing
- Commands for joins, filters, groups
- Nested data types like maps, bags, tuples
- Cannot replace MR in all cases

MAPREDUCE

- Low-level programmatic processing
- Detailed coding is necessary
- Explicit data structures like lists, tuples, maps
- Complex analysis can be coded

SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG RUNNING MODES

Apache Pig

MAPREDUCE MODE

- Runs on Hadoop
- Uses the MR parallelisation
- Works off the HDFS file system
  - Reads of HDFS
  - Writes to HDFS

Command: pig

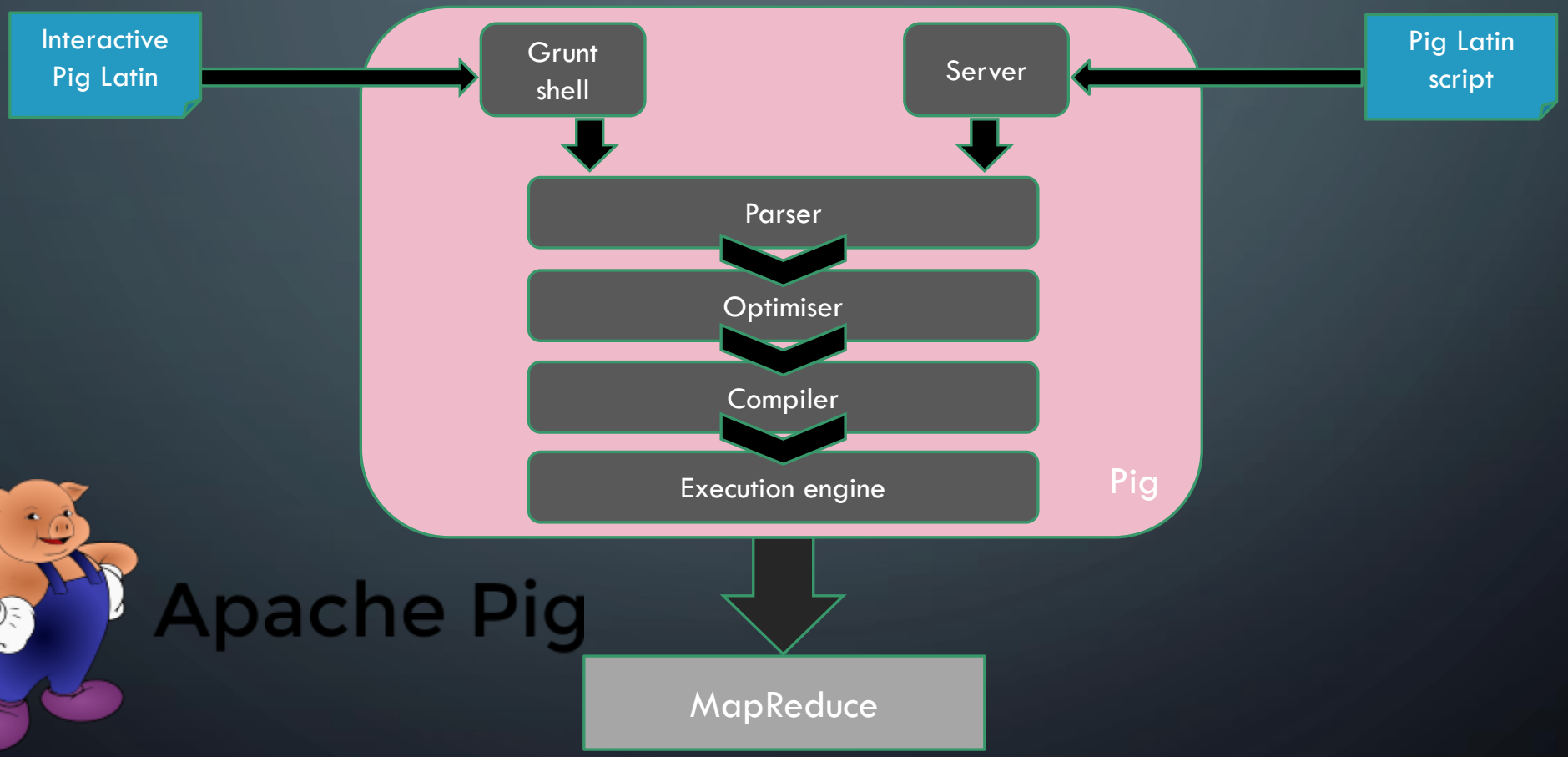
LOCAL MODE

- Runs on a single computer
- Runs a query sequentially
- Works from the local file system
  - Reads one file
  - Writes one file (one reducer)

Command: pig –x local

SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG ARCHITECTURE



Apache Pig

SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG DATA STRUCTURES – TUPLE AND BAG

ID	Name	Street	Phone
1	Bob	Lithcombe	999345
2	Wang	Croyden	890999
3	Srikanth	Kew	433534
4	Veronica	Hawthorn	345490

tuple

bag (several tuples)

Bag representation

{(2, Wang, Croyden, 890999), (3, Srikanth, Kew, 433534), (4, Veronica, Hawthorn, 345490)}

Tuple is pronounced 'tyoople' !

SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG DATA STRUCTURES – MAP

Id # 1,
Name # Bob,
Address # Lithcombe,
Phone # 934582

Id # 2,
Name # Wang,
Address # Croyden,
Phone # 4335424

key # value

Atom in Pig Latin means 'primitive type' like int, char, float, double

SWINBURNE UNIVERSITY OF TECHNOLOGY

EXAMPLE FILE

SalesJan2009.csv

Transaction_date	Product	Price	Payment	Name	City	State
7/8/20 22:00	Product1	1200	Mastercard	Carolina	Basildon	England

SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG LATIN

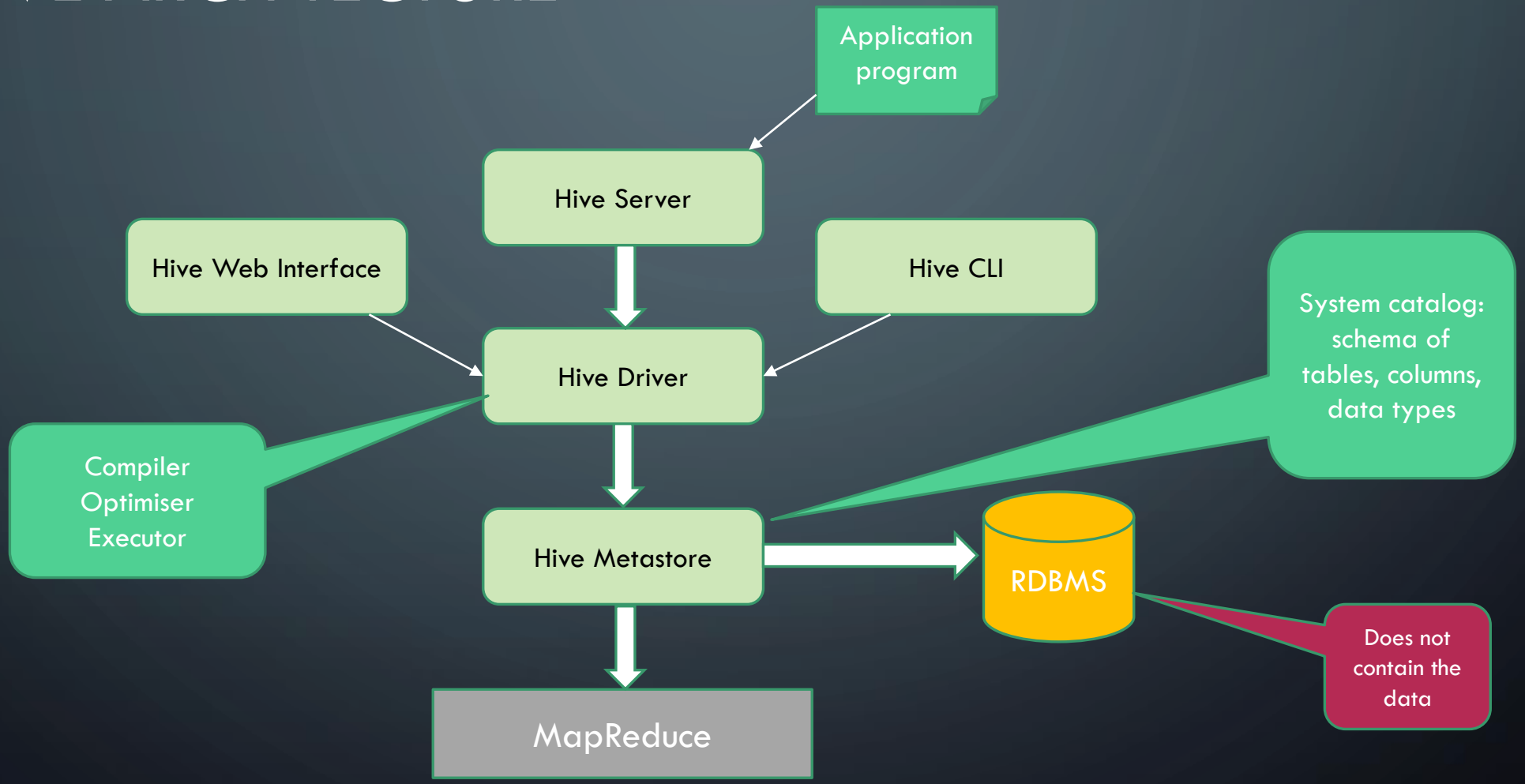

```
rawsales = LOAD ‘xxx.csv' USING PigStorage(',') AS (Transaction_date:chararray, Product:chararray, Price:int, ....);

salesdata = FILTER rawsales BY Price IS NOT NULL;

salesandcountry = FOREACH salesdata GENERATE (chararray) Country AS country, (int) Price AS price;

countrygroups = GROUP salesandcountry BY country;

salespercountry = FOREACH countrygroups GENERATE group as country, SUM(salesandcountry.price) AS totalsales;

sortedsales = ORDER salespercountry BY totalsales desc;

STORE sortedsales INTO '/pigdata/SalesByCountry' USING PigStorage(' ');
```

HIVE

Hadoop Tool for structured data

SWINBURNE UNIVERSITY OF TECHNOLOGY

HIVE ARCHITECTURE



SWINBURNE UNIVERSITY OF TECHNOLOGY

HIVEQL (HQL)

- Includes basic SQL statements:
  - FROM clause & subqueries — SELECT name FROM (SELECT * FROM employee WHERE salary > 5000);
  - JOIN — SELECT name, role FROM employee e JOIN role r ON e.role_id = r.id;
  - multi-table INSERT — Details later – very useful for building an RDB from a .csv file
  - multi-grouping — SELECT SUM(total), year, month, location .... GROUP BY year, month, location;
- AND ALSO
  - TRANSFORM for pluggable scripts — Details later – very useful for changing things on the fly

SWINBURNE UNIVERSITY OF TECHNOLOGY

MULTI-TABLE INSERT - NORMALISING

Sales.csv

Time	Product	Salesrep_id	Salesrep_name	Salesrep_phone	Cust_id	Cust_name
7/8/20 22:00	Trailer	13001	Pete	0993495	200021	Song

Salesrep

Salesrep_id	Salesrep_name	Salesrep_phone
13001	Pete	0993495

Customer

Cust_id	Cust_name
200021	Song

Sale

Time	Product	Salesrep_id	Cust_id
7/8/20 22:00	Trailer	13001	200021

SWINBURNE UNIVERSITY OF TECHNOLOGY

MULTI-TABLE INSERT - HQL

```
FROM Sales.csv s

INSERT OVERWRITE TABLE Sale SELECT s.Time, s.Product, s.Salesrep_id, s.Cust_id
INSERT OVERWRITE TABLE Customer SELECT s.Cust_id, s. Cust_name
INSERT OVERWRITE TABLE Salesrep SELECT s.Salesrep_id, s.Salesrep_name, s.Salesrep_phone;
```


SWINBURNE UNIVERSITY OF TECHNOLOGY

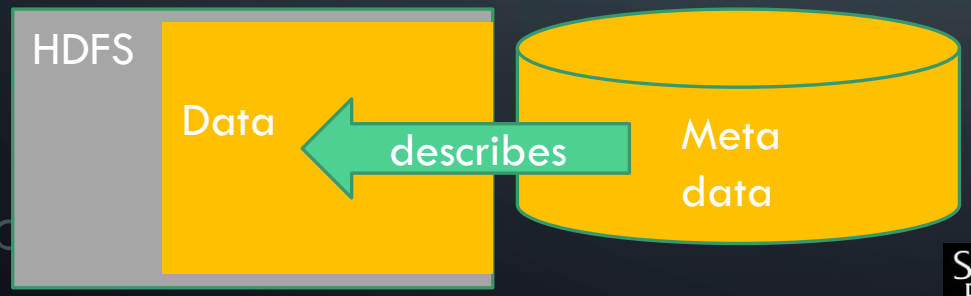
IS HIVE A RELATIONAL DATABASE?

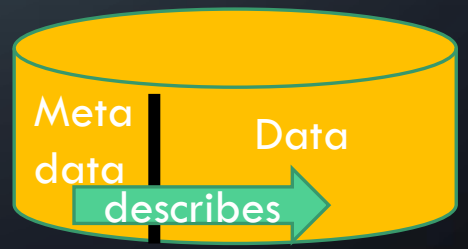
HIVE

- Schema on read
  - Storage in files on HDFS
  - No OLTP (it's a data warehouse!)
- It's for analysis

RDBMS

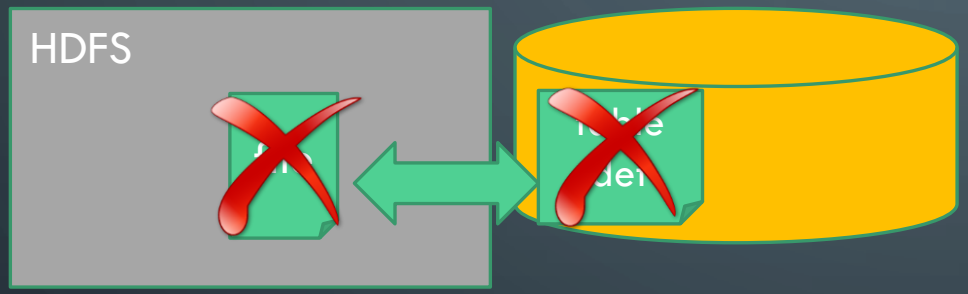
- Schema on write
  - Storage in internal custom file system
  - OLTP
- It's for storage





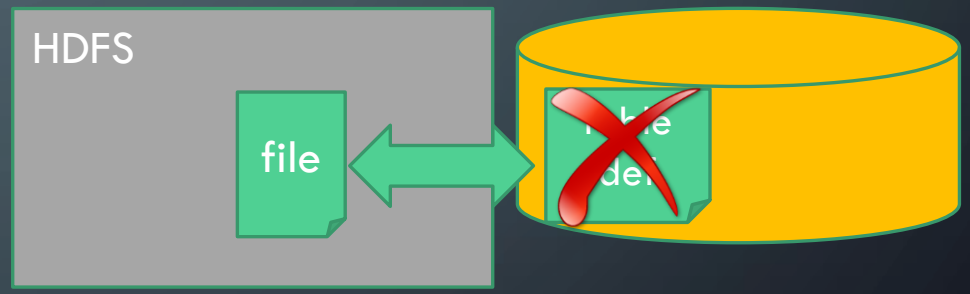
MANAGED AND EXTERNAL TABLES

MANAGED




```
CREATE TABLE <tablename>
(col1 datatype, col2 datatype)
ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t'
STORED as textfile;
```

EXTERNAL




```
CREATE EXTERNAL TABLE <tablename>
(col1 datatype, col2 datatype)
ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t'
STORED as textfile;
```

SWINBURNE UNIVERSITY OF TECHNOLOGY

HIVE LATENCY ISSUES

Part files can still be very large

```
SELECT * FROM sale
WHERE time BETWEEN
   2020-08-01 AND 2020-08-31;
```

/usr/datastore/

- part_00001
- part_00002
- part_00003
- part_00004

Part files containing sales

sale

Time	Product	Salesrep_id	Cust_id
7/8/20 22:00	Trailer	13001	200021

SWINBURNE UNIVERSITY OF TECHNOLOGY

HIVE PARTITIONS

/usr/datastore/time=2020-08-01

- part_00001
- part_00002
- part_00003
- part_00004

/usr/datastore/time=2020-08-02

- part_00001
- part_00002
- part_00003
- part_00004

Store by one column

sale

Time	Product	Salesrep_id	Cust_id
7/8/20 22:00	Trailer	13001	200021

SWINBURNE UNIVERSITY OF TECHNOLOGY

HIVE BUCKETS

/usr/datastore/time=2020-08-02

Secondary column

part_00001	Salesrep 13001, 13002, 13003
part_00002	Salesrep 13004, 13005, 13006
part_00003	Salesrep 13007, 13008, 13009
part_00004	Salesrep 13010, 13011, 13012
part_00005	Salesrep 13013, 13014, 13015

SWINBURNE UNIVERSITY OF TECHNOLOGY

CREATING PARTITIONS AND BUCKETS

```
CREATE TABLE sales
PARTITION BY (time)
CLUSTERED BY (salesrep_id) INTO 5 BUCKETS
```


SWINBURNE UNIVERSITY OF TECHNOLOGY

PIG - HIVE

PIG

- Procedural language
  - For programmers
- Executed on the client
- Inspired by SQL

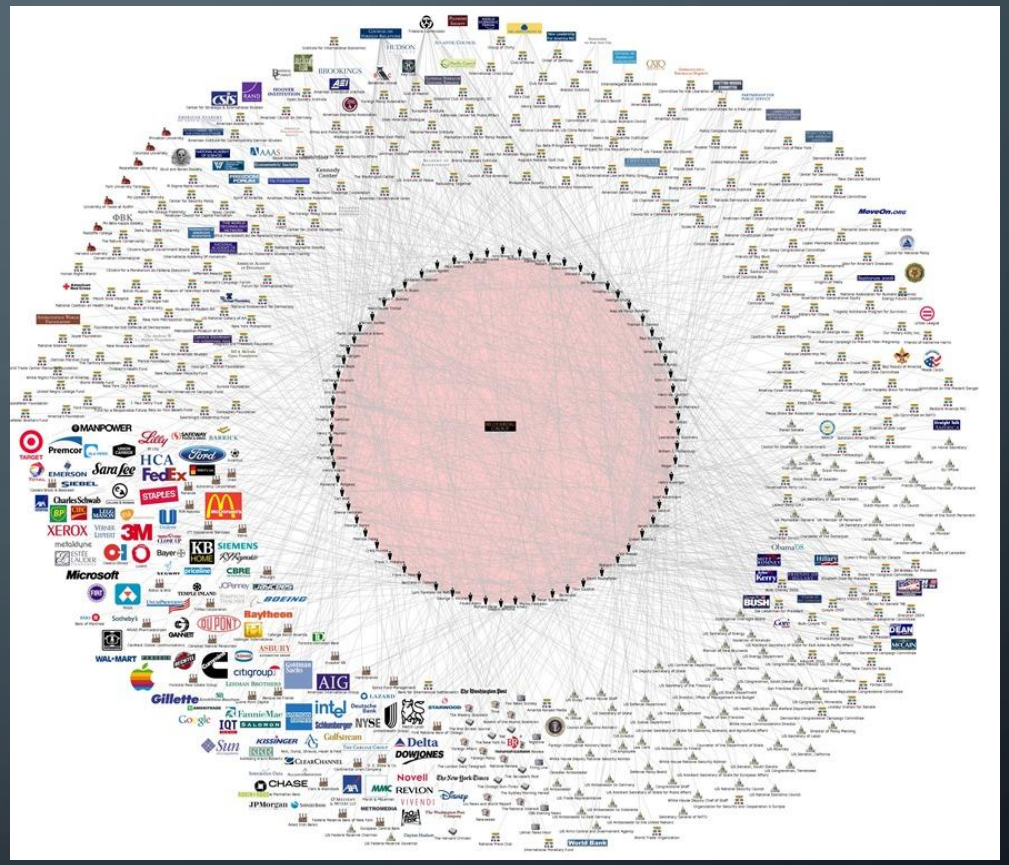
HIVE

- Declarative language
  - For data scientists
- Executed on the server
- Based on SQL

SWINBURNE UNIVERSITY OF TECHNOLOGY

THE BIG PICTURE

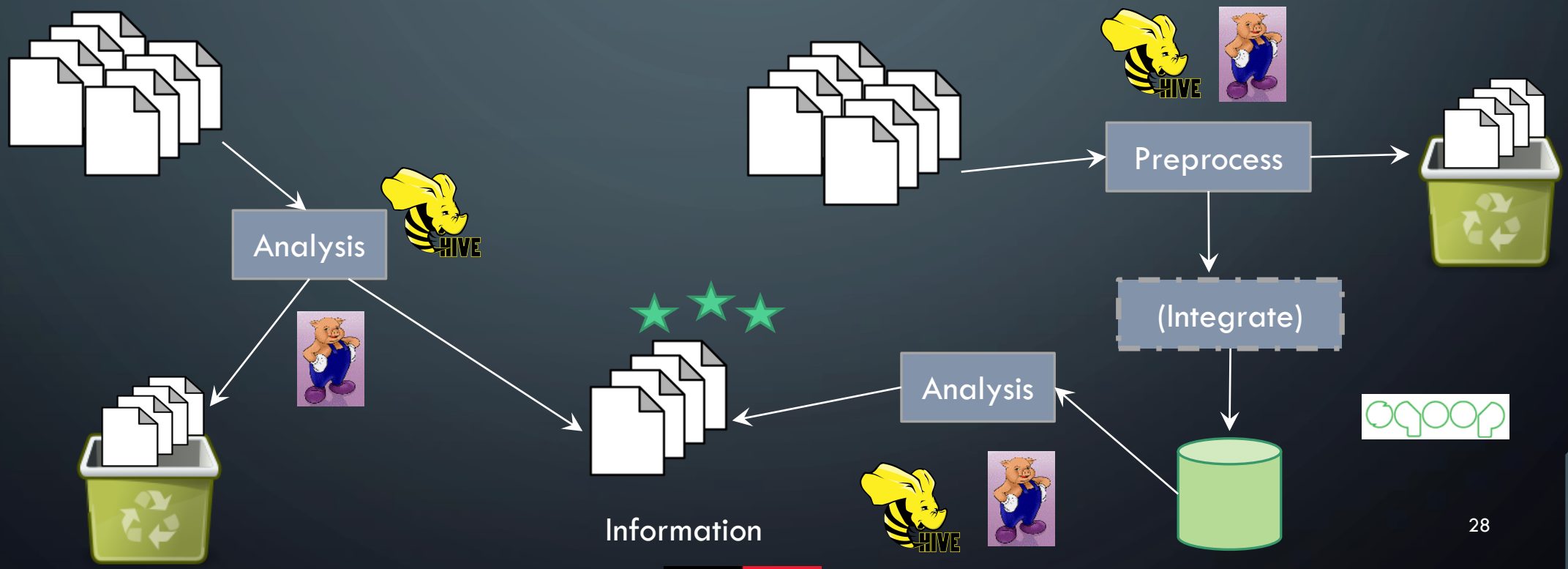
Data processing in the big data age



SWINBURNE UNIVERSITY OF TECHNOLOGY

HOW DO WE DEAL WITH BIG DATA?

- Access only once
- Access several times



SUMMARY

- Hadoop and Apache tools provide the means to process big data sets in a versatile way.
- Pig and Hive are information retrieval tools that are easy to use.
- Parellelisaton and distribution are the underlying principle of all tools.
- Information retrieval can be part of the preprocessing or transformation phase.
  - Big data processing is not rigid ETL.

SWINBURNE UNIVERSITY OF TECHNOLOGY